

A Bilinear Isochronous Oscillator: Global Action–Angle Coordinates and Quantum Interface Spectrum

Source-local theorem reconstruction

2026-09-03

Abstract

We close the classical and quantum dynamics of a joined pair of quadratic half-wells. Every nonzero classical orbit has one energy-independent least period exactly when both stiffnesses are positive. Its global action–angle chart is C^1 and piecewise analytic, but detects unequal stiffness at second order. The natural Friedrichs quantization has simple discrete spectrum characterized completely by a parabolic-cylinder interface Wronskian. Flat-side, free, zero-energy, and equal-frequency boundaries are kept explicit.

1 The common-period owner

For $x_+ = \max\{x, 0\}$, $x_- = \min\{x, 0\}$, and $\omega_+, \omega_- \geq 0$, consider

$$H(x, p) = \frac{p^2}{2} + V(x), \quad V(x) = \frac{\omega_+^2 x_+^2 + \omega_-^2 x_-^2}{2}. \quad (1)$$

The force is continuous and globally Lipschitz, hence the flow is unique and complete, and H is conserved.

Theorem 1 (all-energy isochrony). *Every nonzero orbit of (1) is bounded and periodic with a common least period if and only if $\omega_+, \omega_- > 0$. In that chamber,*

$$T = \pi \left(\frac{1}{\omega_+} + \frac{1}{\omega_-} \right), \quad J(E) = \frac{E}{2} \left(\frac{1}{\omega_+} + \frac{1}{\omega_-} \right), \quad \Omega = \frac{dE}{dJ} = \frac{2}{1/\omega_+ + 1/\omega_-}. \quad (2)$$

Proof. At $E > 0$, write $B = \sqrt{2E}$ and start at $(0, B)$. The right half-excursion is

$$x = \frac{B}{\omega_+} \sin(\omega_+ t), \quad p = B \cos(\omega_+ t), \quad 0 \leq t \leq \frac{\pi}{\omega_+}.$$

Starting next from $(0, -B)$, with $s = t - \pi/\omega_+$, the left one is

$$x = -\frac{B}{\omega_-} \sin(\omega_- s), \quad p = -B \cos(\omega_- s), \quad 0 \leq s \leq \frac{\pi}{\omega_-}.$$

They give the least period in (2). Their two half-ellipse areas are $\pi E/\omega_+$ and $\pi E/\omega_-$, proving the action formula. If a frequency vanishes, an orbit entering that flat half-axis with nonzero speed escapes linearly; if both vanish the motion is free. This proves the necessity. $\square$

2 Seam-compatible action–angle coordinates

Put $\theta_+ = \pi\Omega/\omega_+$, $B = \sqrt{2\Omega J}$, and measure $\theta = \Omega t$ from $(0, p > 0)$. The inverse chart is

$$\begin{aligned} 0 \leq \theta \leq \theta_+ : \quad x &= \frac{B}{\omega_+} \sin \frac{\omega_+ \theta}{\Omega}, \quad p = B \cos \frac{\omega_+ \theta}{\Omega}, \\ \theta_+ \leq \theta \leq 2\pi : \quad x &= -\frac{B}{\omega_-} \sin \frac{\omega_- (\theta - \theta_+)}{\Omega}, \quad p = -B \cos \frac{\omega_- (\theta - \theta_+)}{\Omega}. \end{aligned} \quad (3)$$

Values and first J, θ derivatives agree at the two seams. On each open piece,

$$x_\theta p_J - x_J p_\theta = 1,$$

so (3) is a global C^1 , piecewise-analytic symplectic chart from $(0, \infty) \times \mathbb{R}/(2\pi\mathbb{Z})$ to the punctured plane, with $dx \wedge dp = d\theta \wedge dJ$ and $\dot{\theta} = \Omega$. At θ_+ , the one-sided values of $p_{\theta\theta}$ are $B(\omega_+/\Omega)^2$ and $B(\omega_-/\Omega)^2$; the chart is not C^2 unless the frequencies agree. Each normalized seam-to-seam half-flow matrix is $-I_2$. Thus the common-period map and its derivative are exactly I_2 . This is the seam-compatible action–angle owner.

3 Friedrichs quantization and interface Wronskian

Let $\mathcal{H}$ be the Friedrichs operator of

$$\mathfrak{q}[\psi] = \frac{1}{2} \int_{\mathbb{R}} |\psi'|^2 dx + \int_{\mathbb{R}} V |\psi|^2 dx.$$

For positive frequencies, $V(x) \geq \min(\omega_+^2, \omega_-^2)x^2/2$. Rellich compactness locally and this tail bound make the form-domain embedding compact. Hence $\mathcal{H}$ is self-adjoint with compact resolvent.

Theorem 2 (complete quantum matching). *Put $\nu_{\pm}(\lambda) = \lambda/\omega_{\pm} - 1/2$. The spectrum of $\mathcal{H}$ consists of simple real eigenvalues tending to infinity, and $\lambda \in \mathbb{R}$ is an eigenvalue if and only if*

$$\sqrt{\omega_+} D'_{\nu_+}(0) D_{\nu_-}(0) + \sqrt{\omega_-} D'_{\nu_-}(0) D_{\nu_+}(0) = 0. \quad (4)$$

Proof. Up to scalars, the unique decaying half-line solutions are

$$D_{\nu_+}(\sqrt{2\omega_+} x) \quad (x > 0), \quad D_{\nu_-}(-\sqrt{2\omega_-} x) \quad (x < 0).$$

Operator-domain functions and their first derivatives are continuous at zero. The determinant of these two matching conditions, after cancelling $\sqrt{2}$, is exactly (4), including its plus sign. Thus (4) is necessary and sufficient. Compact self-adjointness supplies the complete discrete spectrum. Two eigenfunctions for one eigenvalue have constant Wronskian, which vanishes at infinity, so they are proportional. $\square$

If $\omega_+ = \omega_- = \omega$, (4) is $2\sqrt{\omega} D_{\nu}(0) D'_{\nu}(0) = 0$. The exact gamma-function values at zero show that its zeros are precisely $\nu = 0, 1, 2, \dots$, recovering $\lambda_n = \omega(n + 1/2)$. Classical isochrony does not assert this ladder when the frequencies differ.

4 Boundaries, evidence, and scope

For positive stiffnesses, $E = 0$ is only the origin and is omitted from the chart. If exactly one stiffness vanishes, the flat half-axis is a continuum of rest equilibria and other classical trajectories escape. Quantum compact resolvent is lost; nonnegativity, flat-side Weyl sequences, and the absence of an L^2 affine or oscillatory flat-side solution show that the spectrum as a set is $[0, \infty)$ with no eigenvalues. If both vanish, this is the free particle. No globally C^∞ seam claim is made.

Exact finite receipts check 60 rational energy/frequency rows, 85 harmonic levels, both half-flow matrices, and all zero-stiffness faces. Analytic arguments, not finite rows, prove the continuum and operator theorems. Dorignac supplies model lineage and the classical/quantum warning [1]; DLMF fixes the Weber convention [2]; Teschl supplies the one-dimensional self-adjoint framework [3]. The formulas and software are independently reconstructed without a priority claim.

The Route-A tuple is

(A0_FAIL, A1_PASS_ANALYTIC, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION);

the overall verdict is rejection. Scope is NO_BAD_EULER_OR_ROOT_NUMBER; Route B is false. The source interface Wronskian is not a target determinant, target-zero match, or Hilbert–Pólya operator. No target local data, Euler factor, root number, automorphy, target divisor, or target functional equation is claimed.

References

- [1] J. Dorignac, *On the quantum spectrum of isochronous potentials*, *J. Phys. A* 38 (2005), 6183–6210. DOI: 10.1088/0305-4470/38/27/007.
- [2] NIST Digital Library of Mathematical Functions, Chapter 12, Section 12.2.
- [3] G. Teschl, *Mathematical Methods in Quantum Mechanics*, 2nd ed., AMS GSM 157, 2014. DOI: 10.1090/gsm/157.